

Johns Hopkins



Smarter monitoring starts at the surface

May 1st , 2026

THE PROBLEM



13.3M

AKI cases globally



68%

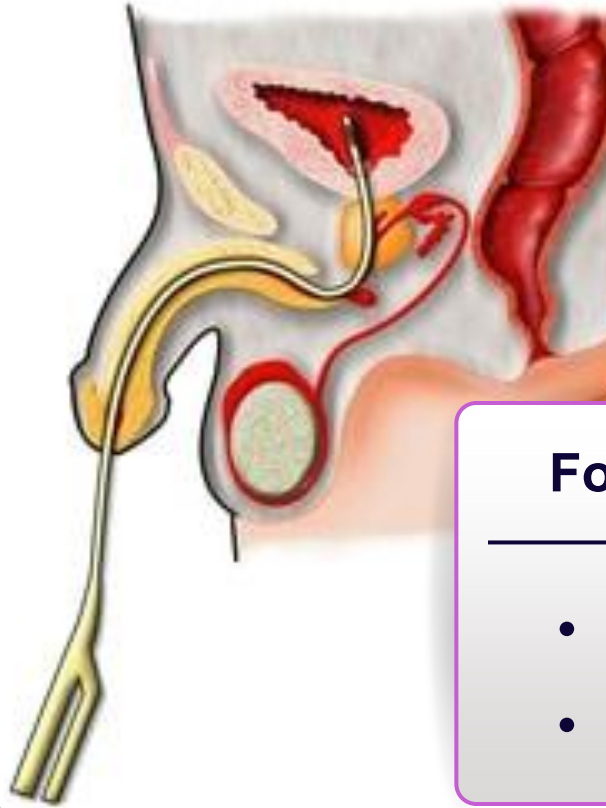
Cases go undetected



9.5B

In hospital costs to U.S

Clinician's current choice



Foley Catheter

- Invasive
- Risk of UTI's

Manual Measurement

- Inefficient
- Costly

Our Solution

1

Belt with 8 disposable electrode patches

2

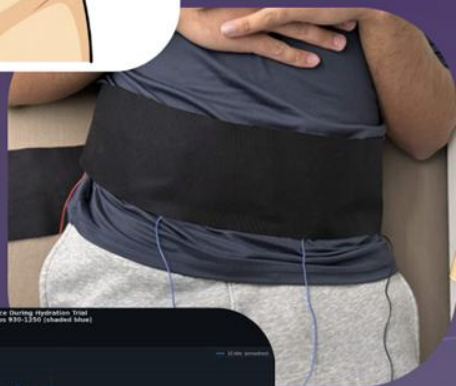
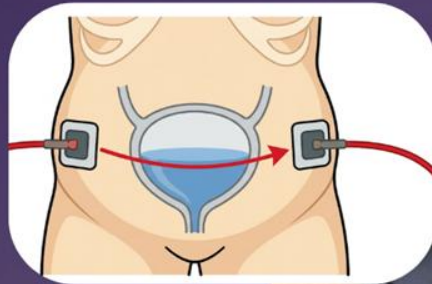
BIA tracks bladder fill via tissue impedance changes

3

Auto-flags when UO < 0.5 mL/kg/hr for 6 hrs

4

Data streams directly to EHR via HL7/FHIR.



Value propositions



Nephrologists

Earlier AKI detection

- Continuous UO data targeting ± 0.075 mL/kg/hr
- KDIGO staging within the critical 6-hour window
- Catch AKI 2–3 hours before creatinine rises



ICU Nurses

Drastic workflow relief

- Eliminates 47-min average charting delay
- >80% reduction in manual measurement tasks
- Setup in <5 min; runs autonomously after



Patients

Safety & comfort

- Zero catheter insertion, zero CAUTI risk
- >90% targeting no-discomfort rating
- Preserve full mobility during recovery



Hospital CMOs

ROI within 24 months

- 1 prevented CAUTI = months of device cost
- Shorter LOS, fewer complications
- Better CMS quality scores & CAUTI metrics

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Competitive Analysis

Solution	Non Invasive	Accurate ($\pm 0.075\text{ml}$)	Continuous	Automated	EHR Ready
Foley Catheter					
Fize Kuo					
Bladder Ultrasound					
Purewick					
Delta P					

Pricing and ROI



Razor Blade Model

\$2000 / device

\$ 200/100 electrodes

- Device sold with 100 free electrode patches
- Long-life hardware → margin improves over lease term
- Electrodes are a high volume consumable = recurring scale revenue

+



SaaS

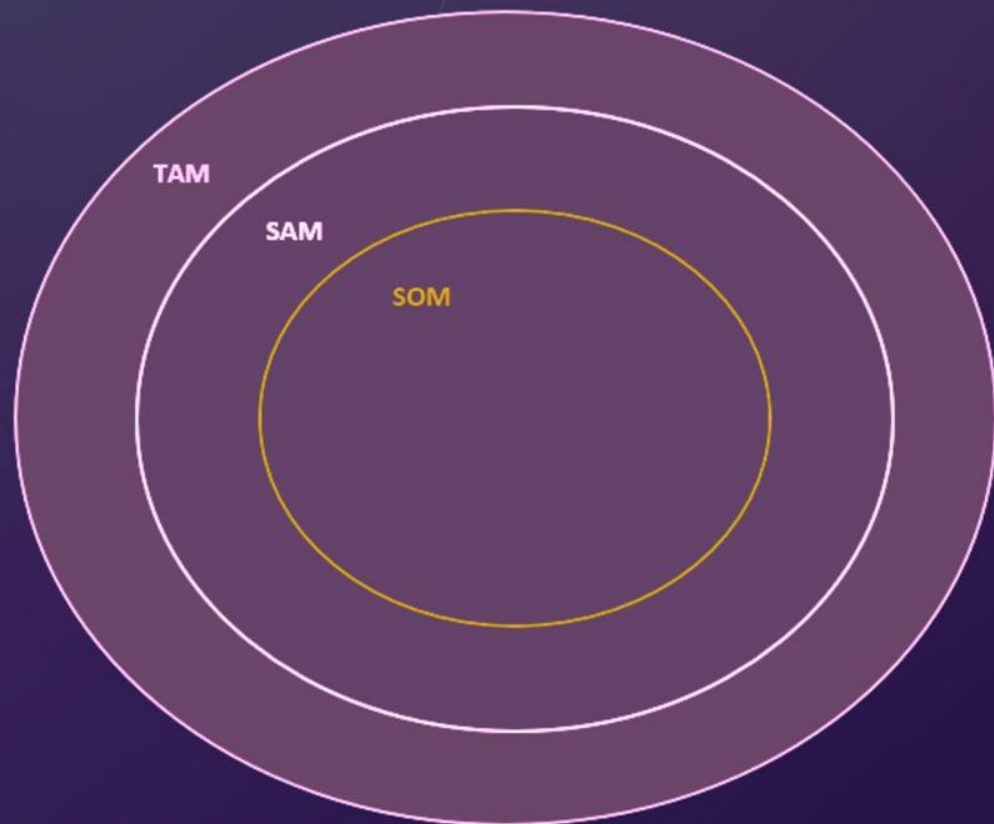
\$100/ patient / year

- Continuous output monitoring and alerting software
- Advanced metrics and measurement history
- Integration with Epic systems

The ROI case:

1 prevented CAUTI saves \$600–\$1,000 (not reimbursed by CMS since 2008). At \$25–40/patch × 15 patches/bed/month, one avoided infection covers the entire month's consumable spend. Hospitals recoup ROI in < 24 months.

Market Opportunity



TAM — Global AKI Treatment Market

\$11B+

SAM — U.S. Inpatient Monitoring Market

\$1.5B

SOM — Medium-Risk ICU Patient

\$150–200M

AKI treatment market growing at 7.3% CAGR.

Our roadmap

2026

MVP Developed

- Silicone electrode belt finalized
- Custom PCB
- 6-hr continuous streaming validated
- Wheatstone bridge amplification

2027

Regulatory

- FDA Pre-Sub (Q-Sub) meeting
- 510(k) submission (Q3 2027)
- First peer-reviewed manuscript
- JHMI observational pilot (ICU)

2029+

Revenue Scale

- \$5–10M annual revenue target
- 10+ anchor health systems
- Home monitoring launch
- International (CE, WHO)

2026–27

Clinical Studies

- IRB approval + human subject pilots
- BMI-stratified testing (3 groups)
- Nurse usability study (mannequin)
- Provisional patent filed

2028

FDA Clearance

- 510(k) cleared (Class II)
- First paying hospital contracts
- Patch manufacturing at scale
- CPT reimbursement application

The Ask

Funds medical grade prototype, further validations, and IRB ready human studies.

60% — University IRB application fee and Clinical trials preparation

\$1200

30% — PCB's, Bladder Ultrasound and prototype improvement materials

\$600

30% — Medical grade electrodes and silicone wearable materials

\$200